

Lab 4: Configuring End devices

Packet Tracer – Configure End Devices

Objectives

Configure Various End Devices in Packet Tracer.

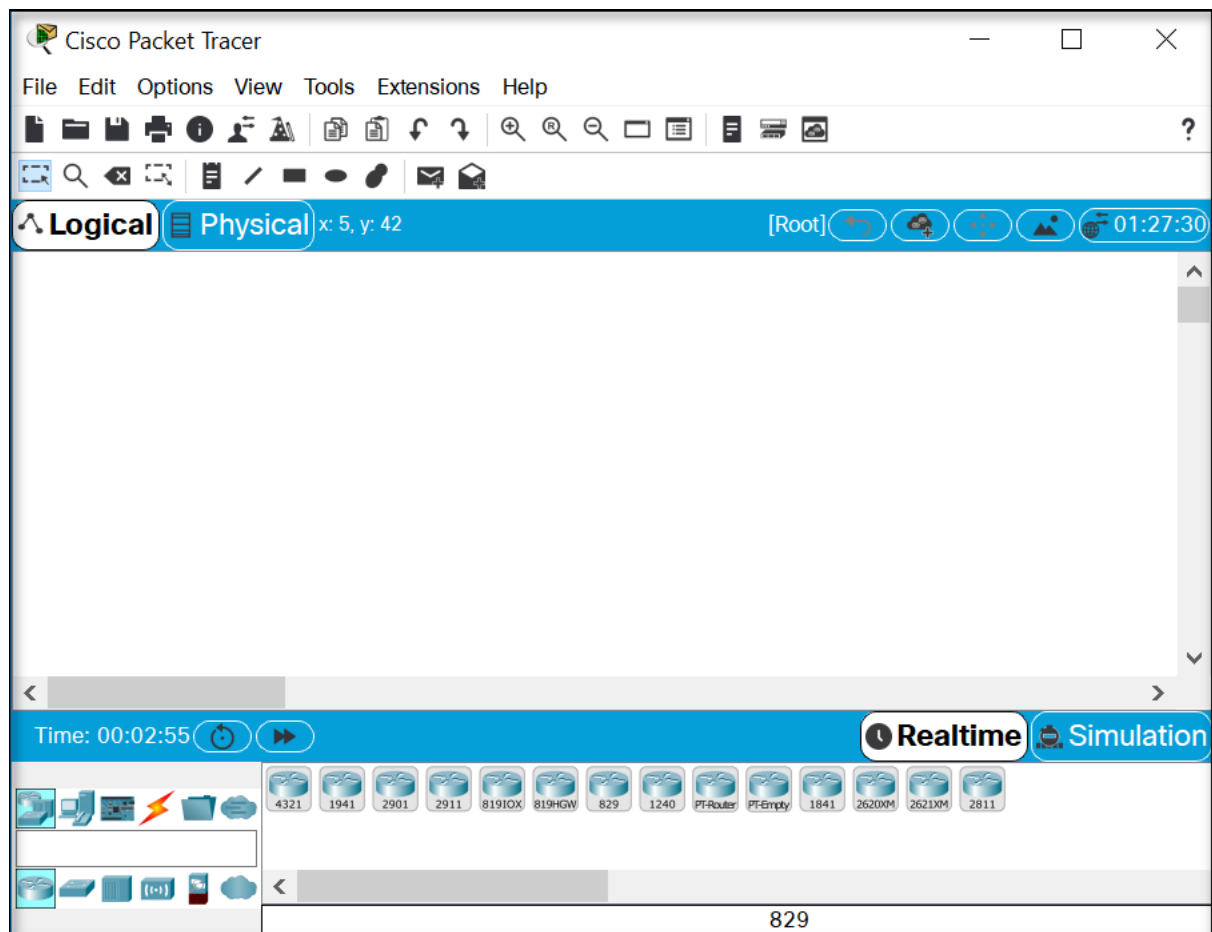
Background / Scenario

In this activity you will construct a simple Packet Tracer network and complete basic configuration of end devices.

Step 1: Launch Packet Tracer.

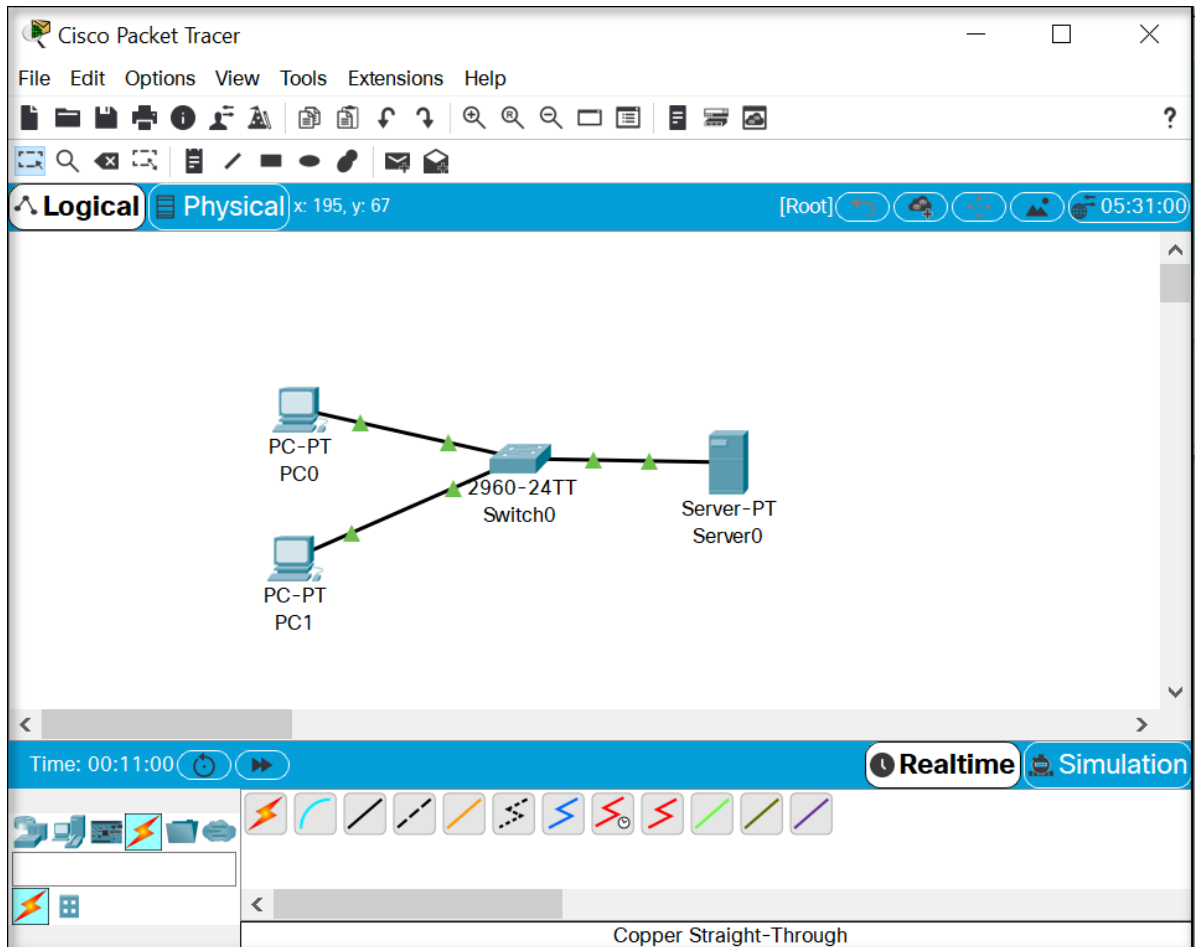
- a. Launch Packet Tracer on your PC or laptop computer

Double click on the **Packet Tracer** icon on your desktop or navigate to the directory that contains the Packet Tracer executable file and launch Packet Tracer. Packet Tracer should open with a blank default **Logical** topology workspace as shown in the figure.



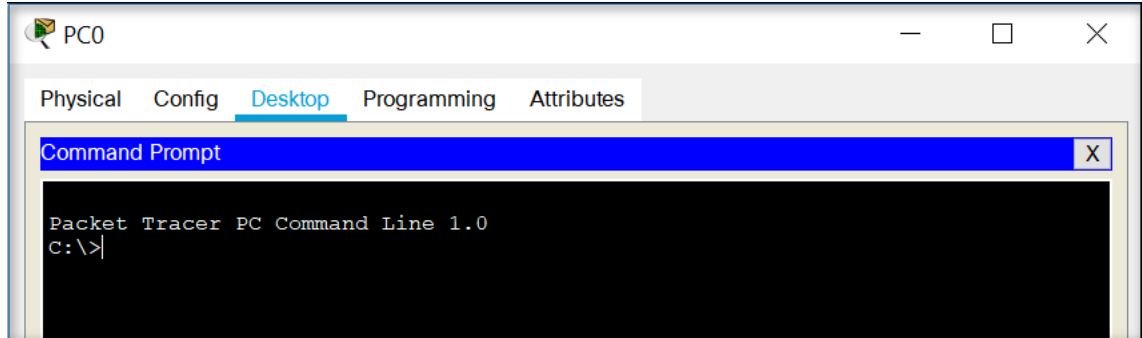
Step 2: Build the topology

1. Create the network shown below (If help is required, please refer to previous activities.).
 - a. Use port FastEthernet0/1 on the switch for PC0
 - b. Use port FastEthernet0/2 on the switch for PC1
 - c. Use port FastEthernet0/3 on the switch for Server0

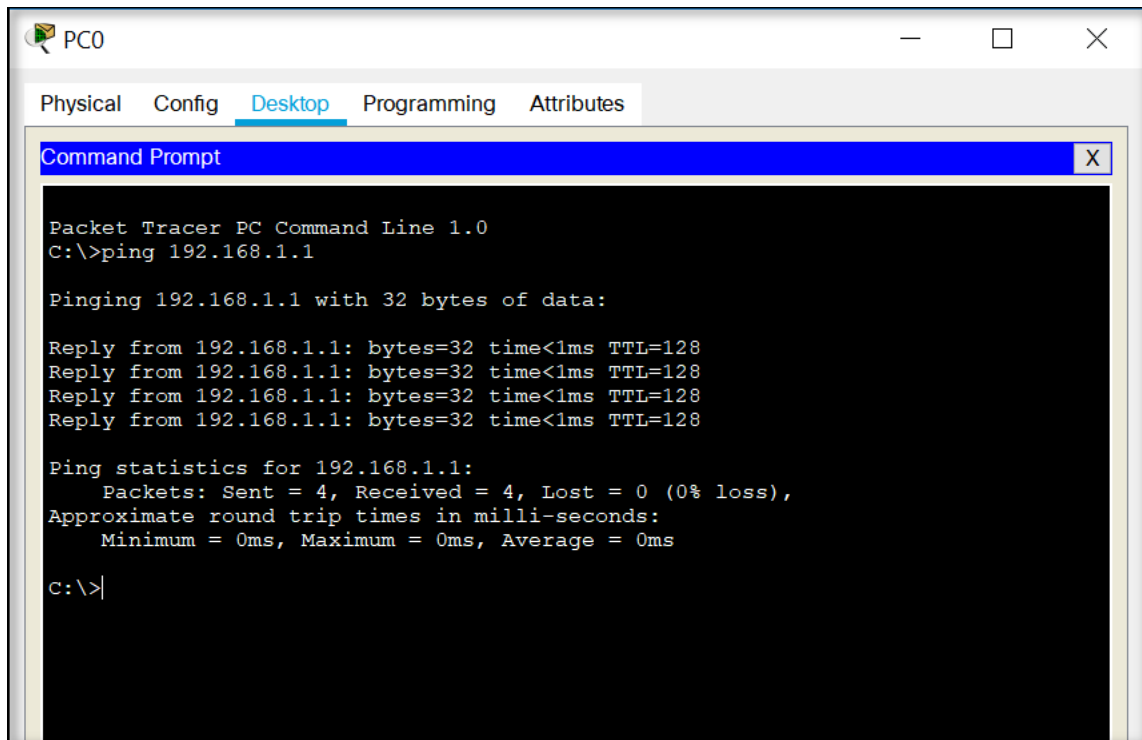


2. Once the link lights all turn green, click on Server0. Then configure it as follows:
 - a. Click on the **Desktop** tab.
 - b. Click on the **IP Configuration** icon.
 - c. Click on the **IP Address** dialog box.
 - d. Type in **192.168.1.1** as the address and press enter.
 - e. A default value of 255.255.255.0 should appear in the **Subnet Mask** field.
 - f. Nothing else in this dialog box needs to be configured, so click the "X" in the upper right corner to close the **IP Configuration** window.
 - g. Click the red "X" in the upper right corner to close the **Server0** window.
3. Click on PC0. Then configure it as follows:

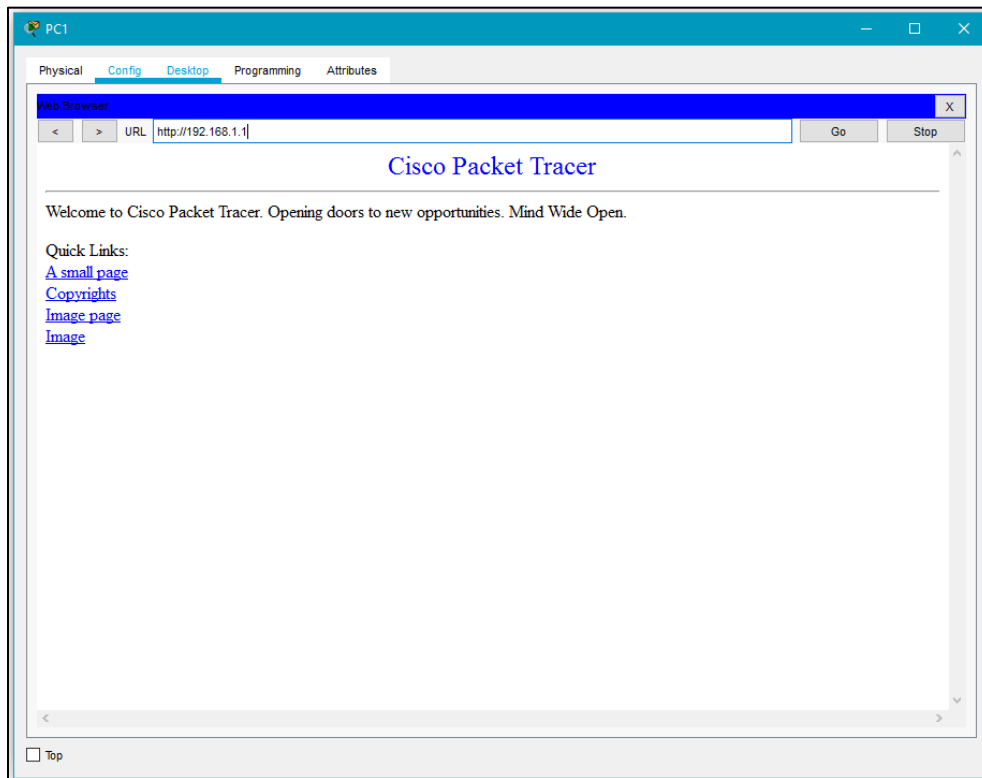
- a. Click on the **Desktop** tab.
- b. Click on the **IP Configuration** icon.
- c. Click on the **IP Address** dialog box.
- d. Type in **192.168.1.2** as the address and press enter.
- e. A default value of 255.255.255.0 should appear in the **Subnet Mask** field.
- f. Nothing else in this dialog box needs to be configured, so click the “X” in the upper right corner to close the **IP Configuration** window
- g. Click on the icon labeled **Command Prompt** and the following prompt should appear:



- h. Type the following command in the prompt: **ping 192.168.1.1** and press enter.
- i. If you have done everything correctly, you should see the following output. Your output could vary a little but the reply statements should be there. If the replies are not there, try redoing the device configuration to this point.



- j. Click the “X” next to the **Command Prompt** title bar.
 - k. Click the red “X” in the upper right corner to close the **PC0** window.
4. Repeat the same configuration and ping steps from #3 on PC1, except use 192.168.1.3 as the IP address. The results should be the same.
5. Finally, click on PC1 again.
- a. Click on the **Desktop** tab, if it is not already open.
 - b. Click on the **Web Browser** icon.
 - c. Type **192.168.1.1** in the URL box and click the [GO] button.
 - d. You should observe the following. If you do not, repeat the earlier steps to confirm the configuration. This happens because the web server feature is on by default in the server and PC1 just connected to the default page.



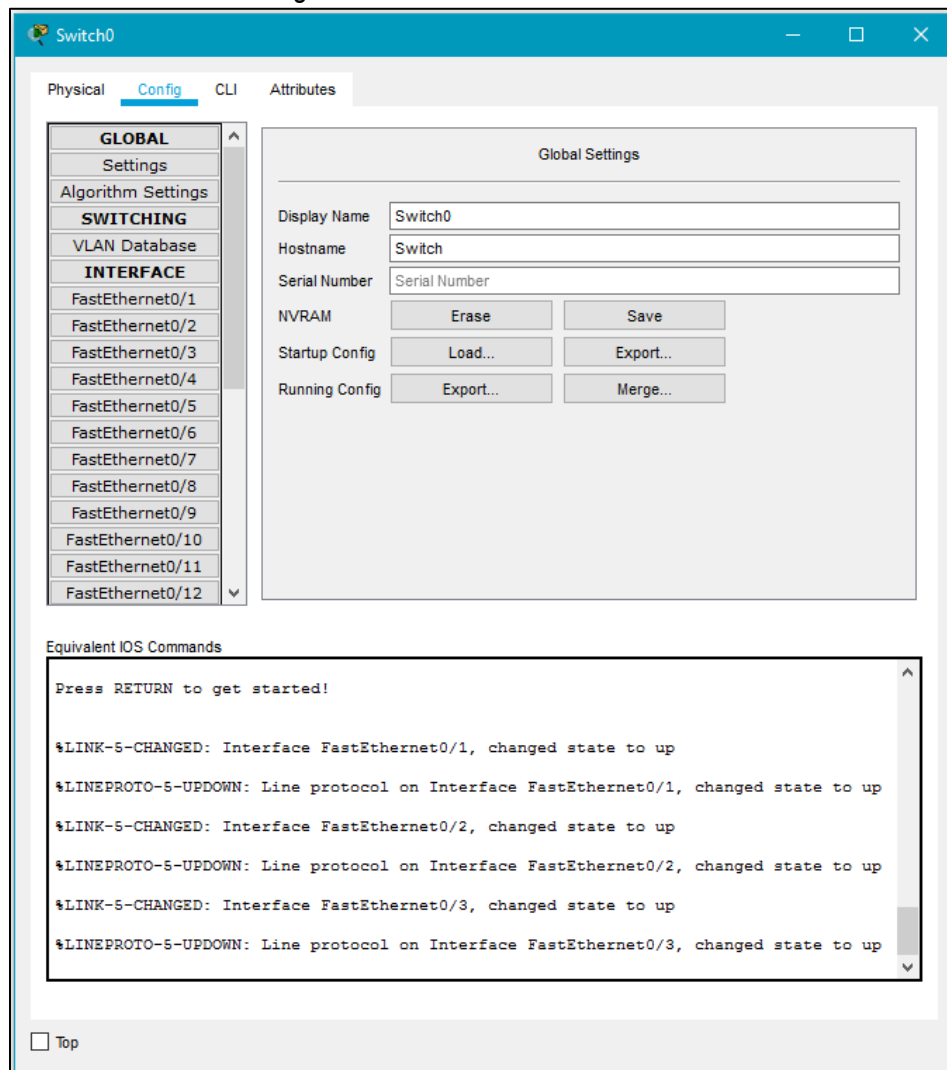
- e. Click on a link and then use the front and back arrows to the left of the URL box to move forward and backward through the pages.
- f. When done, click the “X” next to the **Web Browser** title bar.
- g. Click the red “X” in the upper right corner to close the **PC0** window.

The next section involves some basic configuration of network devices, in this case a switch. Routers have the same tabs as switches so their interface works the same way.

6. Click on Switch0, then click on the **Config** tab.

Note: Previously, a warning about not using the **Config** tab was given because it is not available on real networking equipment, but we are explaining this tab for two reasons.

- i. Some simple devices only have config tabs.
 - ii. The config tab can be useful for basic learning of commands, especially for beginners.
- a. Clicking on the **Config** tab shows a list of components that can be configured on this device. We are not going to cover what these components are, as that is learned in a networking course, but we will show how to navigate and use the interface.



- b. The **Global Settings** tab allows a user to change the name of a device that displays in the workspace. It also allows for changing the internal name shown at the command line prompt as well as buttons for saving, loading, exporting, and erasing configuration files.

- c. Double click in the **Hostname** dialog box highlighting the word Switch, type **Central** and press enter. Packet Tracer will display the IOS commands necessary to accomplish the name change in the **Equivalent IOS Commands** box. The commands displayed should be as follows:

```
Switch>enable
```

```
Switch#configure terminal
```

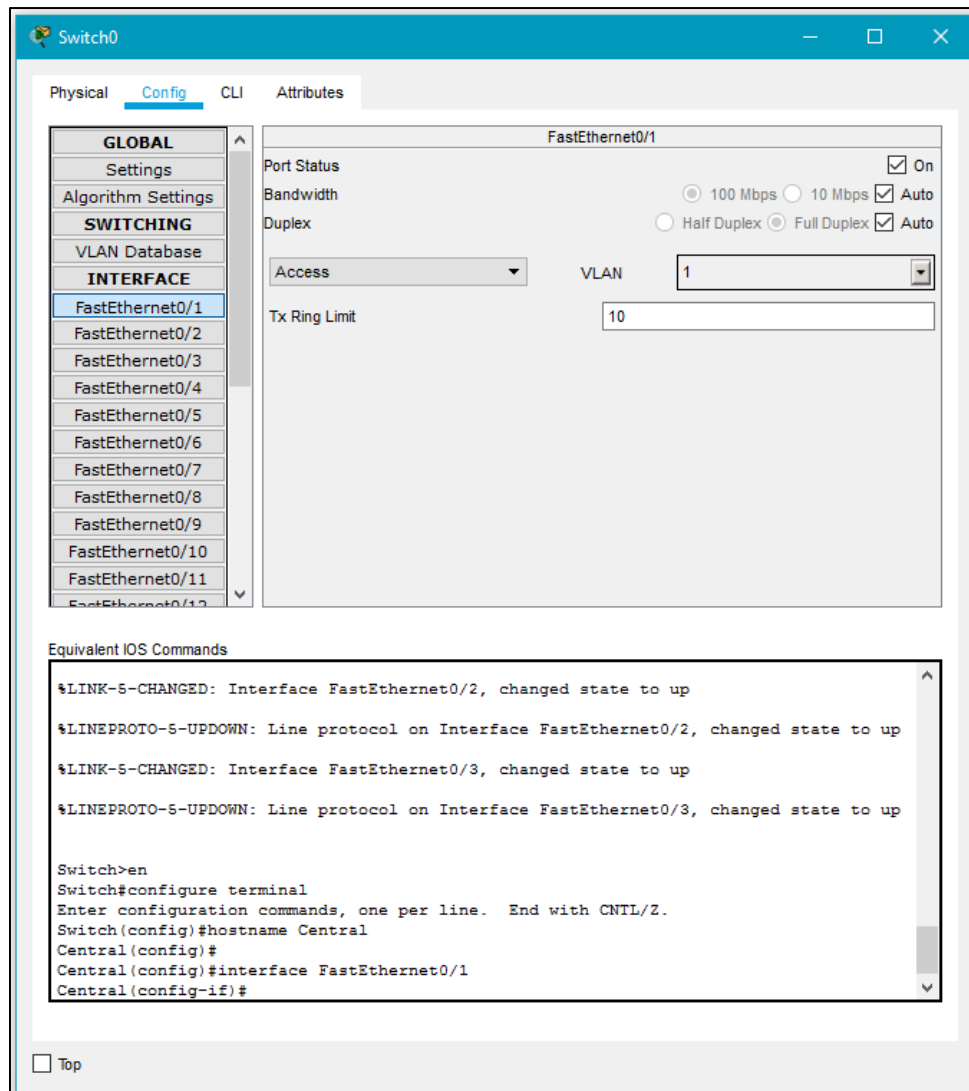
Enter configuration commands, one per line. End with CNTL/Z.

```
Switch(config)#hostname Central
```

```
Central(config)#
```

These would be the commands that would be entered to do the same thing from the command line interface or CLI. If you didn't know how to do this from the CLI, the **Config** tab would show the commands to illustrate how it should be done.

- d. Clicking on the **FastEthernet0/1** label will bring up an Ethernet interface to be configured.
- e. Notice the **Equivalent IOS Commands** box below. It shows a command of "**interface FastEthernet0/1**" which would have been the command used to select the interface from the CLI.



- f. Select the **CLI** tab to switch to the CLI interface. Notice that the same commands that were in the **Equivalent IOS Commands** box are listed in the CLI window.
- g. Click right beside the command prompt at the bottom of the list that looks like this:
"Central(config-if)#"
- h. Then type **shutdown** , and press enter twice

```
Central(config-if)#shutdown
Central(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to
administratively down
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1,
changed state to down
Central(config-if)#
```

This command just shut the interface down from the command line.

- i. Click the red **"X"** in the upper right corner to close the **Server CLI** window. Notice how the link lights for the connection between PC0 and Switch0 are red. Because the interface on the switch was shut down, the connection is no longer active and shows red.

This covers some basic configuration and operation of end devices in Packet Tracer. Please save and close the activity, then exit Packet Tracer.

Lab Tasks

In-Lab Task 1:

- Implement all steps mentioned in lab manual.
- Write all steps implemented in lab in summarized form.

Post-Lab Task 1:

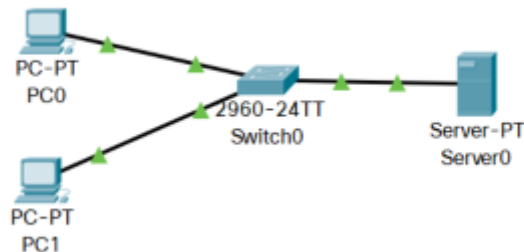


Figure 1: Network

Part a: Design and implement following network in packet tracer and apply the following configurations:

1. Assign IP address and subnet mask to server.
2. Assign IP address and subnet mask to PC0.
3. Assign IP address and subnet mask to PC1.
4. From PC0 ping server and PC1.
5. Access cisco server web browser page.

Use following IP addresses and subnet masks to all devices:

Device	Interface	IP address	Subnet Mask
PC0	FastEthernet0/1	192.168.10.15	255.255.255.0
PC1	FastEthernet0/2	192.168.10.25	255.255.255.0
Server	FastEthernet0/5	192.168.10.35	255.255.255.0

Part b: Apply the following configuration to switch:

1. Replace switch name with your own name
2. Turn off Interface FastEthernet0/1 and FastEthernet0/2 through command in CLI.

Note: Attach screenshots of each step.